MTH5000 MASTERS RESEARCH PROJECT | ORAL PRESENTATION

Machine Learning for Forecasting Hazardous Air Quality Days in Indian Cities

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MONASH University

THE FINDING IN TWO NUMBERS

0.379 to 0.881

hit rate at one day ahead as the alarm budget moves from 40 to 100 alarms a year

about 5 points

the most that changing the model moves it, at a fixed budget

MOTIVATION

A forecast is only useful if it arrives before the hazardous day does

Health basis. PM2.5 exposure is linked to cardiopulmonary and all-cause mortality (Pope and Dockery 2006; Chen and Hoek 2020).

The event. A day above 121 µg/m³, the boundary of the CPCB National Air Quality Index 'Very Poor' category (24-hour mean).

The gap. Two prior Delhi machine learning studies (Masood and Ahmad 2020; Singh and Srivastava 2025) report R^2, RMSE and MAE. Neither evaluates forecasts as a warning system with an explicit alarm budget.

The risk. A model with good average error can still miss the extreme days that are the only ones anyone cares about.

29.7%

of days exceed 121 µg/m³ (920 of 3,093)

1 to 3

days ahead: the forecast horizons

9.8 yrs

of daily data, Nov 2016 to Aug 2026

12

model specifications compared

Exceedances are a minority class, so plain accuracy is never reported: a model that never alarms scores high by construction.

MOTIVATION

Three research questions, and the third turned out to dominate the other two

RQ1 ACCURACY

How accurately can daily concentrations be forecast 1 to 3 days ahead by machine learning, relative to classical and naive benchmarks?

HEADLINE ANSWER

ARIMA beats every ML regressor at every horizon (rMAE 0.940 at h = 1). The deep networks beat it only at h = 1.

RQ2 TWO ARMS

Is it better to forecast the concentration and then threshold it, or to classify the exceedance event directly?

HEADLINE ANSWER

Neither dominates: 21 / 11 / 4 across 36 matched comparisons. Forecast-then-threshold preferred on flexibility.

RQ3 WARNING SYSTEM

As a warning system, what hit rate and false alarm rate does each approach achieve, and how does the decision threshold trade them off?

HEADLINE ANSWER

Budget sweep moves hit rate 0.379 to 0.881 (h = 1). Model choice moves it about 5 points.

DATA

One well-characterised Delhi monitor, nearly ten years of daily PM2.5

Analysis dataset after cleaning	
Source	OpenAQ location 8118, sensor 23534, AirNow
Period	2016-11-09 to 2026-08-27
Calendar days	3,579
Usable daily means	3,093 (86.4% of calendar)
Longest gap	34 consecutive days
Mean, median, maximum	102.7, 71.7, 896.0 µg/m^3
Days above 121	920 (29.7%)

Report Table 3.1.

Why station 8118, and what it costs

- Five stations within 25 km compared on coverage, not nominal record length.
- Every CPCB station serves data only from 19 Feb 2025 (about 18 months), despite metadata saying 2016.
- Only 8118 returns the full record.

Cost: 8118 is an AirNow feed, almost certainly a US diplomatic post monitor. Results describe one site, not Delhi as a whole.

Two defects in the source data were found and corrected before any model was fitted

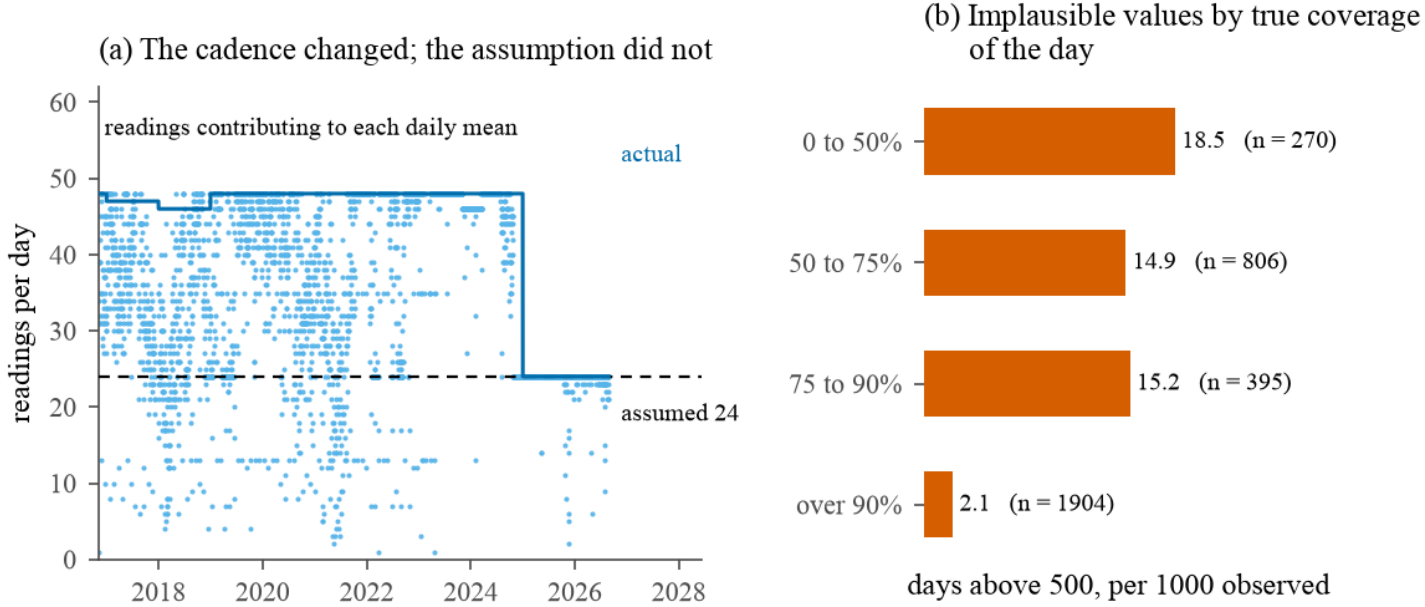


Figure 3.1. (a) readings per daily mean; the fixed denominator of 24 (dashed) against the cadence estimated from the data (solid). (b) implausible values by true coverage.

Rules applied (Table 3.3): value below 0 removes 5 days; coverage below 50% removes 270; value above 1000 removes 7. Total 282 of 3,375. The exceedance rate moves only from 30.0 to 27.0% across every candidate coverage threshold, so the filter is not removing the events of interest.

1. A clipped ceiling

1990 appears three times identically, one of them on 2 September, in the monsoon, between neighbours of 42 and 178. Five negative values as well.

2. A wrong coverage denominator

Completeness assumed 24 readings a day, but the sensor was half-hourly until 2024. A half-day of data counted as complete. Implausible values were about 7 times as frequent at 50 to 75% true coverage as above 90%.

DATA

Exceedances arrive as multi-day winter episodes, and that is the structure worth forecasting

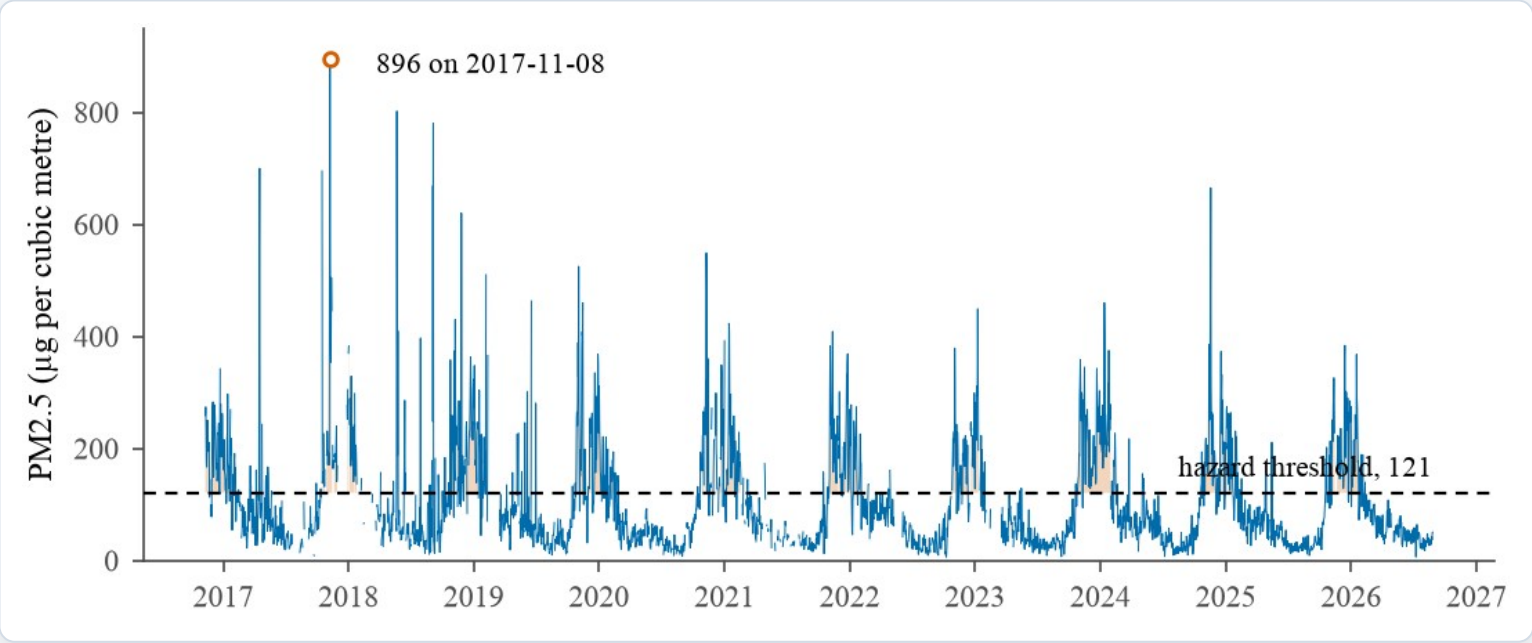


Figure 3.2. The cleaned daily series, with days exceeding the threshold shaded. Maximum 896.0 on 2017-11-08.

102.7
mean µg/m³

71.7
median

2.17
skewness

8.31
excess kurtosis

Fitted in logarithms: right skew, and annual Fourier terms explain 69% of the log variance.

DATA

An assumption from the approved proposal fails: there is no weekly cycle at this station

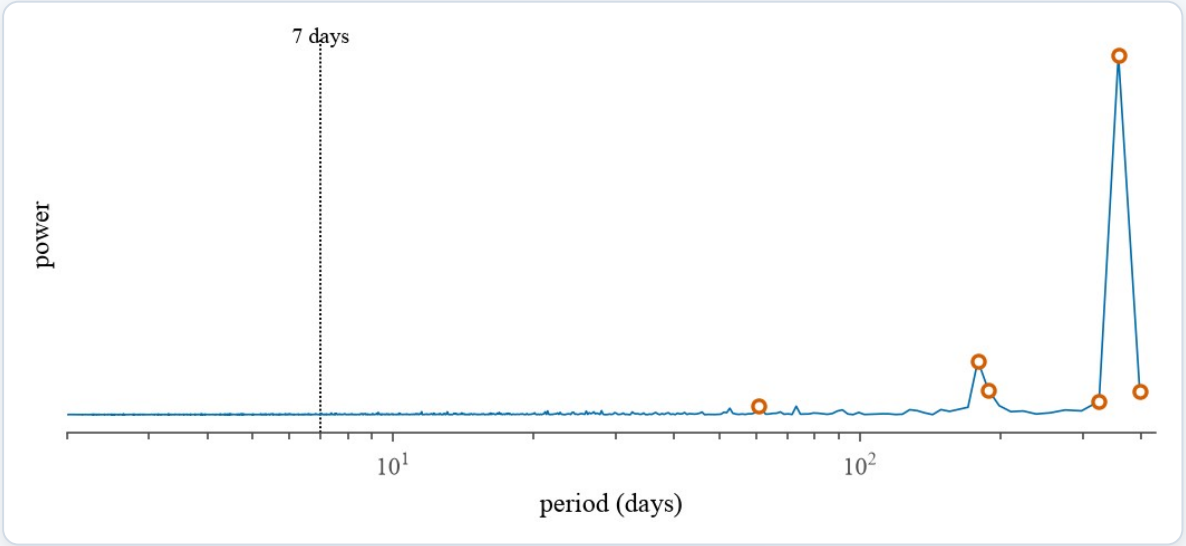


Figure 3.3. Periodogram: the six dominant periods are marked open. Seven days (dotted) sits on the noise floor.

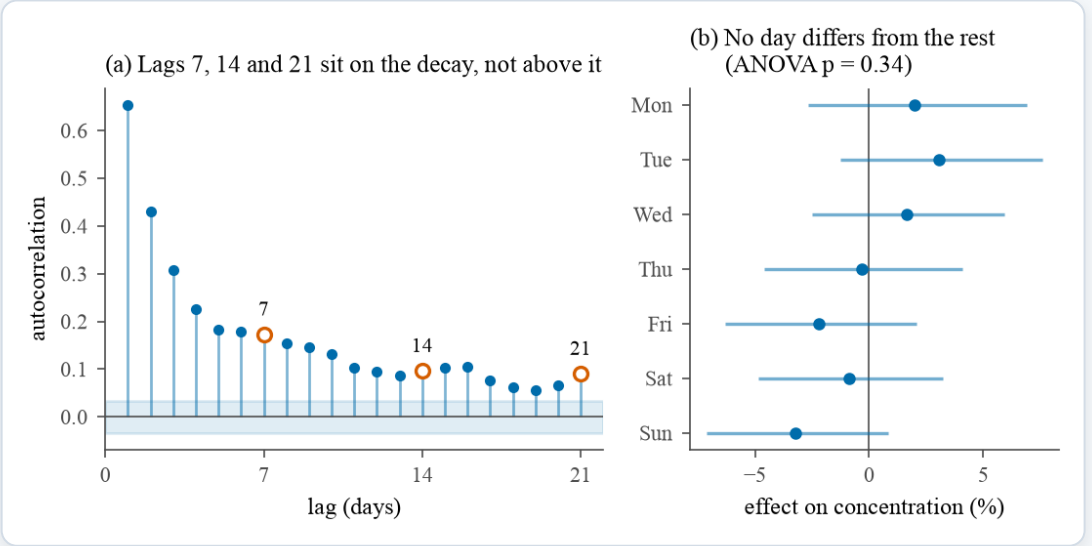


Figure 3.4. Lags 7, 14, 21 lie on the decay curve; every day-of-week interval contains zero.

Periodogram

Dominant periods 357.9, 178.9, 188.4, 397.7, 325.4, 60.7 days. Nothing near 7.

Autocorrelation

Lag 7 = 0.173, lag 6 = 0.179, lag 8 = 0.153: on the decay curve, not above it.

Day of week

6.5% spread. ANOVA F = 1.13, p = 0.34. Kruskal-Wallis p = 0.77.

METHODS

The annual cycle is unmistakable, so ARIMA carries it with Fourier terms and no weekly order

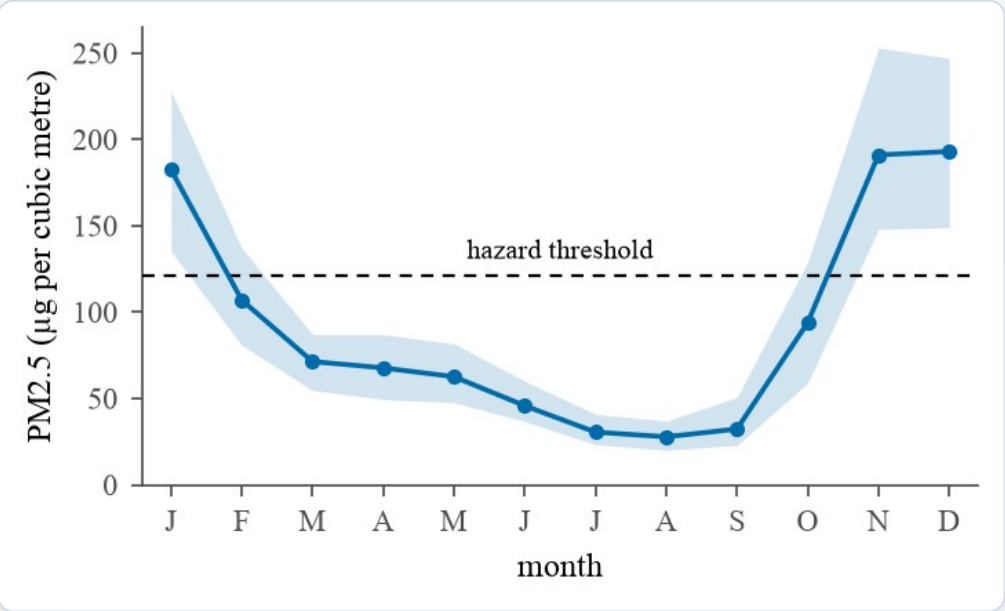


Figure 3.5. Monthly median with interquartile range shaded.

$$(1 - \phi_1 B)(1 - B)\, y_t = c + (1 + \theta_1 B)\, \varepsilon_t + \Sigma_{k=1}^{4} [\, \alpha_k \sin(2\pi k t/365.25) + \beta_k \cos(2\pi k t/365.25) \,]$$

y_t = log(PM2.5), B is the backshift operator, K = 4 annual Fourier pairs, no weekly seasonal order.

Order by AIC over a 24-candidate (p, d, q) grid: (1,1,1) scores 1892.5, next (1,1,2) at 1893.6, a gap of 1.1.

Fitted: $\phi_1 = 0.581$ and $\theta_1 = -0.987$, which is 0.013 inside the invertibility boundary.

Back-transform: median = exp(m), mean = exp(m + $s^2/2$). Both are reported as separate models.

METHODS

Twelve specifications from three families, all scored by one protocol

Classical and naive

- Persistence (repeat today)
- Climatology
- Seasonal naive, lag 7
- ARIMA(1,1,1) + annual Fourier terms

Machine learning

- Ridge regression
- Random forest: level and change target
- Histogram gradient boosting: level and change target

Deep learning (10 seeds each)

- GRU
- LSTM
- Transformer encoder
- Kolmogorov-Arnold-LSTM: bounded check only

Common to all: fitted to log PM2.5, forecast at horizons h = 1, 2, 3 days, refitted as the training window expands, scored on identical forecast origins by identical code.

METHODS

Three machine learning regressors, each controlling variance by a different mechanism

Ridge regression

SHRINKAGE

$$\hat{\beta} = (X^{T}X + \lambda I)^{-1} X^{T}y$$

- Closed form: $X^{T}X + \lambda I$ is positive definite for any $\lambda > 0$, so a unique solution exists even with collinear lags.
- λ chosen by cross-validation at every refit.

Random forest

BAGGING

$$\operatorname{Var} \hat{f}(x) = \rho\sigma^{2} + (1 - \rho)\sigma^{2} / B$$

- Average of B trees on bootstrap resamples.
- Random feature subsets at each split lower the correlation ρ, which is the floor averaging alone cannot beat.

Histogram gradient boosting

ADDITIVE CORRECTION

$$F_{m} = F_{m-1} + \nu\, h_{m}, \quad r_{im} = -\partial L/\partial F$$

- Each tree fits the pseudo-residual: gradient descent in function space.
- Binned features accept missing values natively, so nothing is imputed.

Trees were fitted to both the level of log concentration and its change from the forecast origin: a tree cannot extrapolate beyond the targets it was trained on, and this series drifts across the decade.

METHODS

Three networks under one protocol, so the comparison is the cell and nothing else

Architecture	Size	Optimiser	Parameters
GRU	24 hidden units	Adam, lr = 0.01	2,329
LSTM	24 hidden units	Adam, lr = 0.01	3,097
Transformer	d_model = 16, 2 heads, 1 layer	Adam, lr = 0.001	3,409
Kolmogorov-Arnold-LSTM	24 hidden units, grid size 5	Adam, lr = 0.01	3,288

Report Table 3.6. Parameters are for the six-channel network, confirmed against the fitted model objects.

Shared input: last 30 days, six channels: log PM2.5, missingness flag, two annual Fourier pairs.

Shared training: target is the change in log concentration; Adam on absolute error; early stopping on the latest 15% of each window.

Reported as a range over ten random seeds, never a single favourable one.

Gates

GRU: 2, no cell state

LSTM: 3, plus cell state

Transformer: none, self-attention

Missing input

forward-fill plus a missingness flag, for all three

METHODS

Rolling-origin evaluation: 2,303 forecasts, none of which can see the future



Invariant: at origin t a model may use observations up to and including day t, and nothing else. No future values, no full-series scaling.

Tested, not assumed: corrupt the series from a chosen date, rerun, and require that no earlier forecast changes.

Label arithmetic: the label on row t′ is the value at t′ + h, so the training set ends at t − h. One model per horizon.

Why not one train-test split?

0.903 → **0.940**

single block | rolling origin

ARIMA rMAE at one day ahead. The single block was calmer than the training period and flattered every model.

METHODS

Scored as a warning system: an alarm budget in alarms per year, not just an error

ACCURACY

$$\text{rMAE} = \text{MAE(model)} / \text{MAE(persistence)}$$

1.000 is persistence; below 1 beats it. Identical rows, per horizon, never averaged.

HIT RATE

$$\text{TP} / (\text{TP} + \text{FN})$$

Share of actual hazardous days the model catches.

FALSE ALARM RATE

$$\text{FP} / (\text{FP} + \text{TN})$$

False alarms divided by quiet days, not by alarms raised.

ALARMS PER YEAR

$$365.25\ (\text{TP} + \text{FP}) / n$$

The unit an operator budgets in.

Significance: Diebold-Mariano test, absolute-error loss, Harvey et al. small-sample correction, run once per seed against the same ARIMA forecast. Plain classification accuracy is never reported: with hazardous days a minority, true negatives dominate it.

RESULTS

RQ1: every machine learning regressor loses to ARIMA, at every horizon

Model	h = 1	h = 2	h = 3	Bias, h = 1
GRU, 10 seeds	**0.910 to 0.939**	0.878 to 0.906	0.831 to 0.879	−2.8
LSTM, 10 seeds	**0.913 to 0.932**	0.874 to 0.892	0.824 to 0.847	−2.4
Transformer, 10 seeds	**0.914 to 0.925**	0.870 to 0.893	0.825 to 0.842	−2.8
ARIMA(1,1,1) + Fourier	**0.940**	**0.883**	**0.832**	**−6.4**
Random forest, change	0.968	0.919	0.897	+2.1
Gradient boosting, change	0.983	0.937	0.878	+2.0
Random forest, level	0.988	0.934	0.879	+4.1
Gradient boosting, level	1.008	0.945	0.885	+4.0
Ridge regression	0.997	0.952	0.897	+2.6
Persistence	1.000	1.000	1.000	−0.1
Climatology	1.415	1.060	0.947	+9.4
Seasonal naive, lag 7	1.759	1.325	1.168	+0.1

Report Table 4.1: rMAE against persistence over 2,303 origins. Lower is better. Shaded: ARIMA and the networks' one-day range.

0.940

ARIMA at h = 1. Networks: 0.910 to 0.939

0.831 | 0.832

best network seed vs ARIMA at h = 3: a tie

5 of 5

ML regressors behind ARIMA at every horizon

RESULTS

What the one-day-ahead forecasts look like against the series they predict

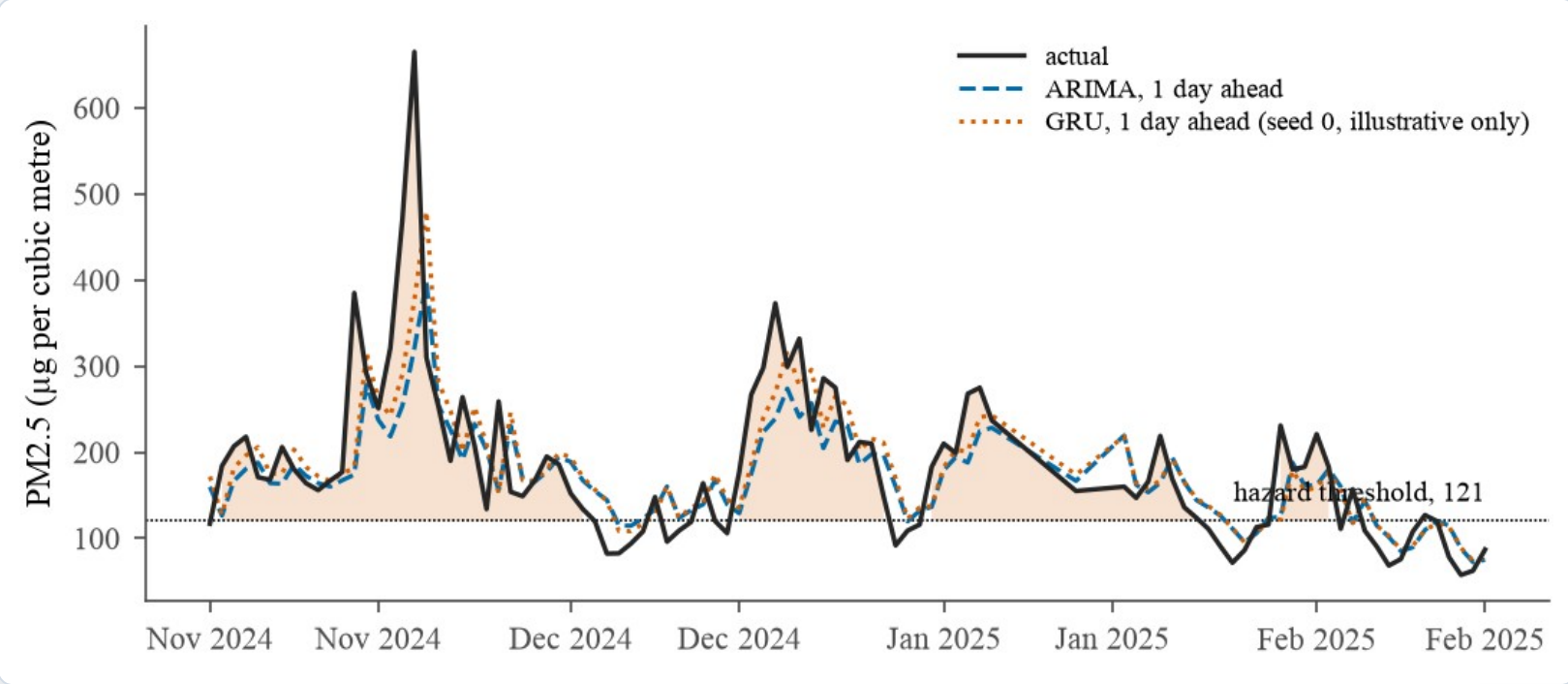


Figure 4.2. One representative winter (November 2024 to February 2025). The GRU line is seed 0 alone, for illustration only.

Both models track the shape of each episode, not only its level. The gap is visible mainly on the sharpest rises, where a short-range nonlinear model would be expected to earn its edge.

RESULTS

A network's result is a range, not a number, and the spread is as large as many of the gaps

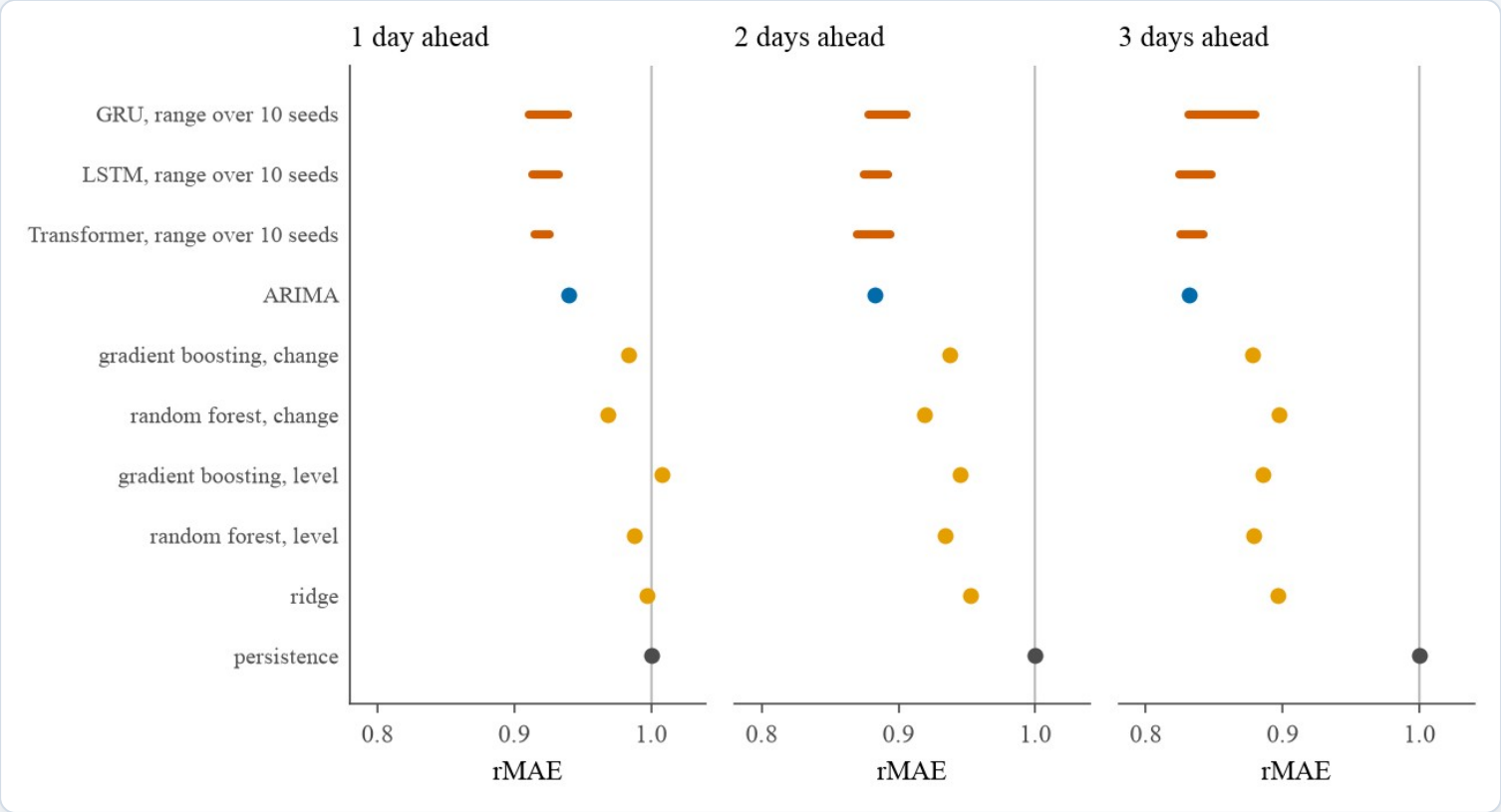


Figure 4.1. Table 4.1 as a dot plot; each network's ten seeds as a range bar. The vertical line at 1.000 is persistence.

GRU spread across ten seeds

0.029 h = 1 **0.028** h = 2

0.048 h = 3: wider than the gap between ARIMA and two other models

Noise floor 0.004: tree models moved this much between two machines (thread-order floating point).

Same seed, same machine still moved the GRU nearly as much as a new seed, so five seeds became ten.

RESULTS

Formal tests are more conservative than counting seeds, and the picture holds

Seeds with p < 0.05 vs ARIMA	h = 1	h = 2	h = 3
GRU	**4 / 10**	0 / 10	1 / 10
GRU with weather	**7 / 10**	0 / 10	1 / 10
LSTM	**5 / 10**	0 / 10	0 / 10
Transformer	**7 / 10**	0 / 10	0 / 10

Report Table 4.2 (Diebold-Mariano, small-sample corrected). The one significant 3-day GRU seed, p = 0.0056, favours ARIMA.

Four of ten seeds significant is itself a small-sample proportion: its 95% Wilson interval runs from 0.17 to 0.69. A different batch of ten could plausibly give a noticeably different count.

14 of 15

regressor tests favour ARIMA at p < 0.05

6 of 27

tests survive a Bonferroni correction

Read beyond one day as suggestive

Twenty-seven tests are run with no correction, so about 1.4 false positives are expected by chance. After Bonferroni, none of the plain GRU's, LSTM's or transformer's best seeds survives.

RESULTS

Two reasons for ruling models out did not survive being tested

ARGUMENT 1: FEWER PARAMETERS FAVOUR THE GRU

LSTM beats GRU at every horizon

by 0.005, 0.008, 0.013 rMAE

3,097 parameters against the GRU's 2,329, a third more. The smaller network is not winning because it is smaller.

ARGUMENT 2: TOO LITTLE DATA FOR A TRANSFORMER

Transformer beats GRU in 8 of 10 seeds

at every horizon, and matches the LSTM

Dr Tian's count of about 4,000 days clears the threshold. At h = 1 it has the strongest formal result of the three: 7 of 10 seeds significant.

Caveat stated in the report: the transformer's four settings were inherited from a single-split check, not tuned. This is one configuration's result.

RESULTS

Weather helps at one day ahead and leaves two and three days a tie

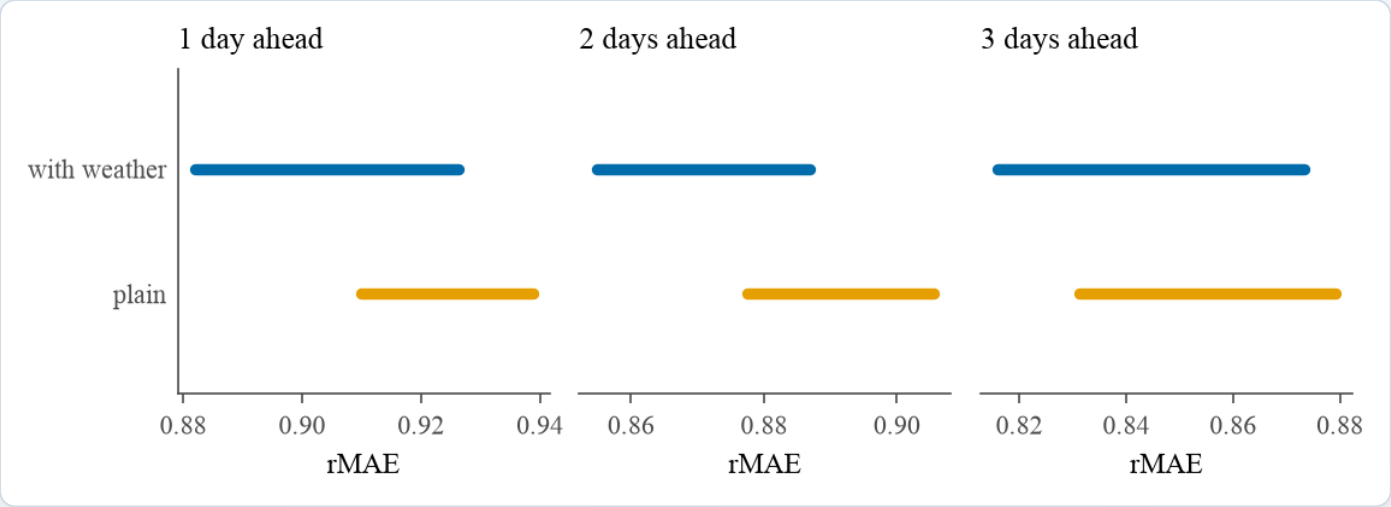


Figure 4.3. The GRU's range over ten seeds, plain against weather-augmented. The bars separate at h = 1 and overlap at h = 2 and 3.

Change in rMAE with weather	h = 1	h = 2	h = 3
Gradient boosting, level	−0.082	−0.020	−0.029
Gradient boosting, change	−0.073	−0.047	+0.006
Ridge	−0.031	−0.005	+0.023
Random forest, level	+0.002	−0.006	+0.053
Random forest, change	−0.003	−0.001	+0.022

Report Table 4.4. Negative is an improvement.

GRU with weather beats ARIMA at h = 1 in all ten seeds (7 of 10 significant). At h = 2, three seeds win, six tie, one loses; at h = 3, three win, two tie, five lose. Under the first five seeds both horizons looked like wins: doubling the seed count is what overturned that. Result survives restriction to origins inside the fully observed weather window.

RESULTS

Carried through the decision threshold, a more accurate forecaster buys about one point of hit rate

Best hit rate, alarms per year	40	60	80	100
h = 1, no weather	0.401	0.609	0.787	0.899
h = 1, weather models added	0.402	0.609	0.792	**0.909**
h = 2, no weather	0.388	0.597	0.771	0.900
h = 2, weather models added	**0.400**	0.597	0.771	0.900
h = 3, no weather	0.383	0.567	0.756	0.876
h = 3, weather models added	0.383	**0.577**	0.756	0.876

Report Table 4.5: best achievable hit rate within an alarm budget, averaged over the ten network seeds, scored on identical origins.

≤ 1.2 points

largest hit-rate gain from adding weather models (h = 2, budget 40)

8.2 points

rMAE gain for gradient boosting at h = 1, which became at most a little over 1 point of hit rate

The ceiling is not the ten original models being too inaccurate: it persists after a materially more accurate forecaster is added.

RESULTS

RQ2: forecast-then-threshold has a real but modest edge over direct classification

h = 1, alarms per year	40	60	80	100
Ridge: forecast then threshold	**0.377**	**0.554**	**0.723**	0.850
Ridge: penalised logistic	0.356	0.550	0.721	**0.856**
Random forest (change): forecast	**0.367**	0.550	**0.729**	**0.867**
Random forest: classifier	0.331	0.550	0.719	0.856
Grad. boosting (change): forecast	**0.379**	**0.554**	0.727	0.854
Grad. boosting: classifier	0.323	0.546	0.727	**0.865**

Report Table 4.6: forecast-then-threshold against direct classification, matched by base algorithm, h = 1. Shaded: the better arm in each column; exact ties unshaded.

21 forecast-then-threshold wins

11 direct classification wins

4 exact ties

of 36 matched comparisons (3 pairs, 3 horizons, 4 budgets)

Median gap 0.011 of hit rate, largest 0.056. It wins 8 of 9 at a 40-alarm budget and 7 of 9 at 60, but loses 6 of 9 at 100.

RESULTS

RQ3: moving the alarm budget changes the hit rate by fifty points, and changing the model by five

Best hit rate, alarms per year	40	60	80	100
Best model, h = 1	**0.379**	**0.567**	**0.735**	**0.881**
Best model, h = 2	0.366	0.538	0.700	0.840
Best model, h = 3	0.359	0.533	0.685	0.827
GRU, 10 seeds, h = 1	0.356–0.377	0.546–0.565	0.719–0.735	0.863–0.881
ARIMA, h = 1	0.358	0.567	0.727	0.867
Gradient boosting (change), h = 1	0.379	0.554	0.727	0.854
Persistence, h = 1	0.371	0.558	0.723	0.873
Climatology, h = 1	0.352	0.521	0.660	0.821
Persistence, h = 3	**0.359**	0.507	0.668	0.791
Climatology, h = 3	0.336	0.510	0.649	0.816

Report Table 4.7. Persistence and climatology at h = 3 are the two models the discussion names.

50 points

0.379 to 0.881: moving the budget, 40 to 100 alarms a year (h = 1)

about 5 points

changing the model at a fixed budget

At h = 3 and 40 alarms a year, persistence (0.359) is the best model in the whole study.

The trade-off curves very nearly coincide, and that is the finding

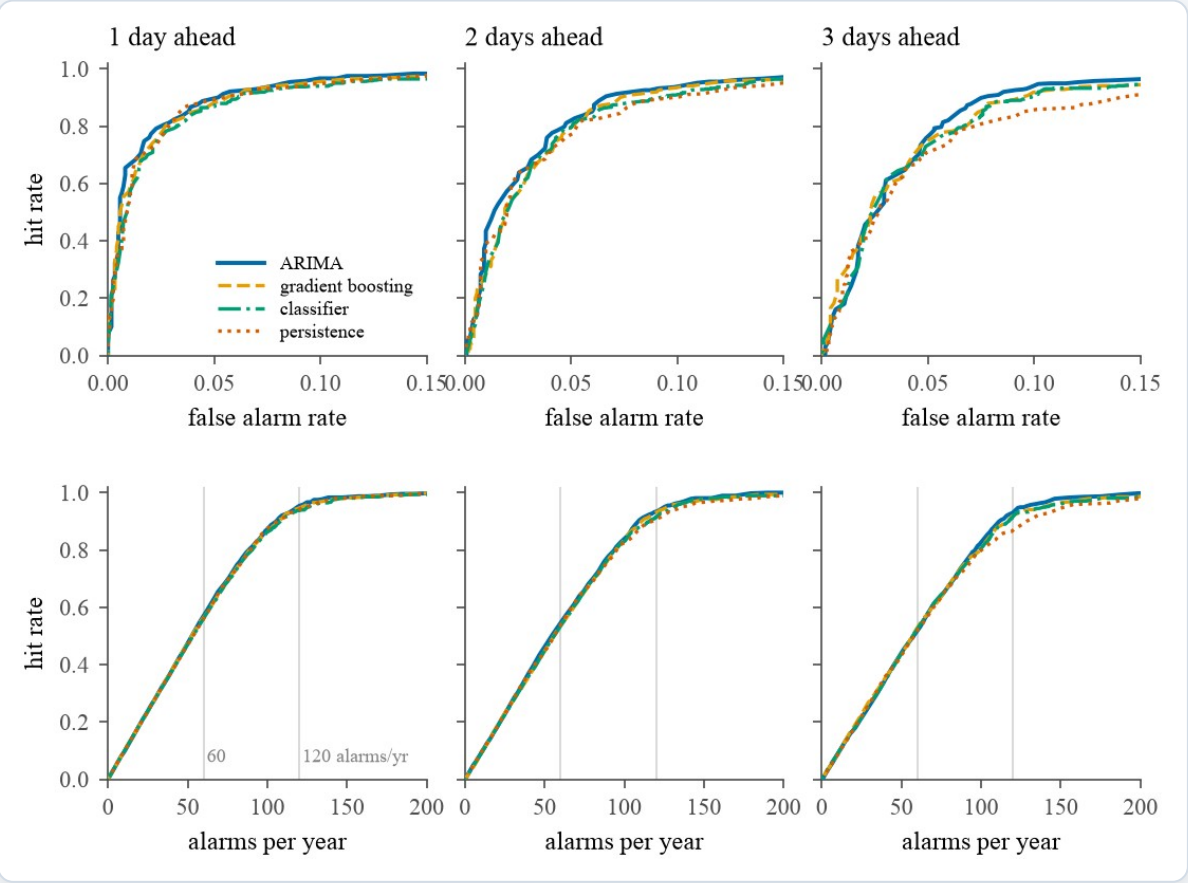


Figure 4.5. Top: hit rate against false alarm rate. Bottom: hit rate against alarms per year. The curves have deliberately not been separated.

Same curve, every model. Each model sits on nearly the same trade-off curve at each horizon.

What differs is where you stand on it, and that is the operator's decision: alarms per year.

Not anti-modelling. ARIMA and the networks earn real, modest gains over persistence (Table 4.1). The leverage in the problem sits elsewhere.

DISCUSSION

The central finding survives every check, and the checks changed what I can claim about the models

Check	What changed	Result
Restore the 7 capped days	Refit full harness on uncapped series	ARIMA 0.935 vs 0.940 (h = 1). Ridge 0.997 → 1.118. GB level 0.883 → 1.006 (h = 3).
Weather coverage gap	Restrict to origins inside observed window	One-day ordering survives; two- and three-day picture unchanged.
Seed count 5 → 10	Doubled for every network	Weather win at h = 2 and 3 did not survive.
GRU window, size (bounded)	6 settings, 3 seeds, one split	Default at or near best at h = 1, 2; worst of six at h = 3 (45-day window 2.2 points better).
GRU Fourier pairs (bounded)	2 pairs → 4 (matches ARIMA)	h = 3 improves 3.96 points, enough to put the mean below ARIMA on this split.
GRU optimiser (bounded)	Adam / RMSprop / SGD, lr ×/÷ 3	h = 3: SGD with momentum about 2 points better (0.839 vs 0.858).

The bounded checks are single-split, three-seed: nothing in the ten-seed rolling-origin Table 4.1 was changed on their strength.

DISCUSSION

What the results do not show

One site, one city

8118 is a diplomatic-enclave monitor. Nothing here is a claim about Delhi as a whole or about other cities.

Observed weather, not forecast weather

A real early-warning system would use numerical weather prediction. Boundary layer height is not available from meteostat.

Scope boundaries

The LSTM and transformer were not given weather channels or the threshold sweep. The KAN-LSTM is a bounded check only.

Small search spaces

Tree grids were small; the ML results are a lower bound. The GRU's 3-day configuration is the one its own checks least defend.

Significance testing

27 tests, no joint correction, best-seed selection, and no bootstrap check of the heavy tails of the loss differential.

Not bit-reproducible

Network training moves at a fixed seed; ten seeds reduce the problem but do not remove it.

CONCLUSION

Three questions asked, three answered

RQ1	**Accuracy vs benchmarks**	ARIMA beats every ML regressor at every horizon (rMAE 0.940 at h = 1). The GRU beats ARIMA only at h = 1, across all ten seeds. LSTM and transformer follow the same pattern.
RQ2	**Forecast then threshold, or classify?**	Neither dominates: 21 / 11 / 4 across 36 matched comparisons. Forecast-then-threshold preferred: a concentration can be rethresholded for any future policy.
RQ3	**Model choice vs decision threshold**	Budget sweep moves hit rate from 0.379 to 0.881 (h = 1, 40 to 100 alarms a year). Model choice moves it about 5 points at a fixed budget.

Along the way: two source-data defects corrected before modelling, and a weekly-cycle assumption from the proposal tested and found false.

CONCLUSION

Design the warning system around its alarm budget, and treat the model as a refinement

RECOMMENDATION

A hazardous air quality warning system should be designed around the alarm budget its operator can justify, with the forecasting model treated as a secondary refinement.

A model comparison that does not fix the alarm budget it was evaluated at has answered a question nobody who operates the system asked.

SIX EXTENSIONS

1. **Forecast weather.** Numerical weather prediction instead of observed conditions.
2. **Boundary layer height.** From a reanalysis product such as ERA5.
3. **Four harmonics for the GRU** under the full ten-seed rolling-origin evaluation.
4. **A second station,** ideally inside the CPCB network.
5. **Complete the LSTM and transformer:** weather channels and the threshold sweep.
6. **Promote the KAN-LSTM** from a bounded check to a ten-seed evaluation.

Thank you. Questions?

The alarm budget a decision maker accepts dominates the choice of model.

Code, data and report: github.com/JC16092/mth5000-aqi-forecast

Backup slides follow: ARIMA order search, significance tables, GRU sensitivity, architecture checks.

BACKUP

B1. The ARIMA order search in full: all 24 candidates by AIC

(p,d,q)	AIC	BIC	note
(1,1,1)	**1892.5**	**1955.8**	**chosen**
(1,1,2)	1893.6	1962.6	
(2,1,1)	1894.4	1963.4	
(3,1,1)	1895.8	1970.5	
(2,1,2)	1895.8	1970.6	no convergence
(3,1,2)	1897.7	1978.2	
(1,0,2)	1910.4	1985.2	no convergence
(2,0,1)	1912.0	1986.7	no convergence
(1,0,1)	1912.1	1981.1	
(2,0,0)	1914.7	1983.7	
(3,0,1)	1915.5	1996.0	no convergence
(3,0,2)	1919.9	2006.1	

(p,d,q)	AIC	BIC	note
(1,0,0)	1927.4	1990.7	
(0,0,2)	2009.2	2078.2	
(0,1,2)	2022.5	2085.7	
(2,0,2)	2100.0	2180.5	no convergence
(3,0,0)	2126.5	2201.2	no convergence
(3,1,0)	2153.8	2222.8	no convergence
(0,0,1)	2167.9	2231.1	
(0,1,1)	2177.9	2235.4	
(2,1,0)	2196.7	2259.9	no convergence
(1,1,0)	2265.2	2322.7	
(0,1,0)	2355.6	2407.4	
(0,0,0)	2734.0	2791.5	

Report Table 3.4, sorted by AIC.

BACKUP

B2. Diebold-Mariano tests against ARIMA

Network vs. ARIMA	h	DM statistic	p-value	Sig. seeds
Plain GRU	1	−3.08 to −0.10	0.002 to 0.923	4/10
Plain GRU	2	−0.47 to 1.73	0.083 to 0.992	0/10
Plain GRU	3	−0.05 to 2.77	0.006 to 0.987	1/10
GRU with weather	1	−3.92 to −0.87	<0.001 to 0.383	7/10
GRU with weather	2	−1.78 to 0.26	0.075 to 0.942	0/10
GRU with weather	3	−0.83 to 1.97	0.049 to 0.907	1/10
LSTM	1	−2.59 to −0.66	0.010 to 0.511	5/10
LSTM	2	−0.65 to 0.83	0.409 to 0.974	0/10
LSTM	3	−0.59 to 1.05	0.292 to 0.995	0/10
Transformer	1	−2.74 to −1.39	0.006 to 0.164	7/10
Transformer	2	−1.34 to 0.84	0.181 to 0.995	0/10
Transformer	3	−0.63 to 0.73	0.463 to 0.988	0/10

ARIMA vs. ML regressor	h = 1	h = 2	h = 3
Ridge	4.25 (<0.001)	3.85 (<0.001)	3.62 (<0.001)
RF, level	3.05 (0.002)	2.43 (0.015)	2.19 (0.029)
RF, change	1.99 (0.046)	**1.93 (0.054)**	3.53 (<0.001)
GB, level	3.92 (<0.001)	2.65 (0.008)	2.26 (0.024)
GB, change	3.00 (0.003)	2.51 (0.012)	1.96 (0.050)

Report Tables 4.2 and 4.3. Positive statistic favours ARIMA. All 15 regressor pairings favour ARIMA; 14 reach $p < 0.05$ (RF change, $h = 2$: $p = 0.054$, shaded). Cells show DM statistic (p-value).

BACKUP

B3. GRU sensitivity: three bounded checks, three seeds, one split

Check	Setting	h = 1	h = 2	h = 3
Architecture	H = 24, 30-day (default)	0.914	0.886	0.858
	H = 12	0.927	0.899	0.856
	H = 48	0.924	0.889	0.850
	15-day window	0.922	0.919	0.839
	45-day window	0.912	0.907	0.836
	60-day window	0.912	0.931	0.851
Seasonal input	K = 4 (matches ARIMA)	0.913	0.881	0.819
Training	Adam, lr = 0.003	0.913	0.888	0.864
	Adam, lr = 0.03	0.933	0.912	0.905
	SGD, lr = 0.01	0.930	0.889	0.839
	RMSprop, lr = 0.01	0.923	0.914	0.882

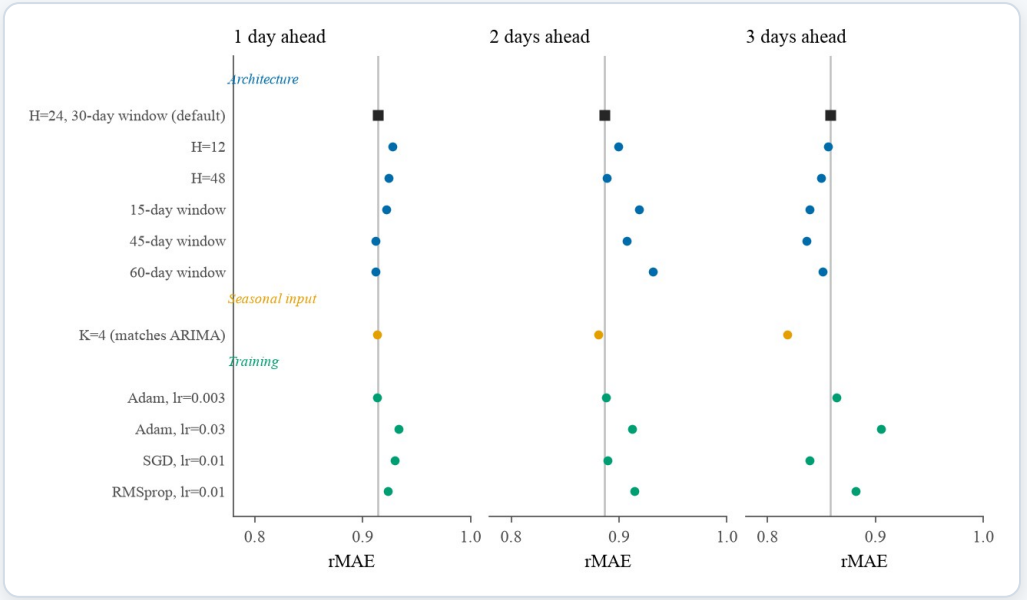


Figure 5.1. Mean rMAE across the same three seeds, one row per setting; the vertical line marks the default.

BACKUP

B4. ARIMA coefficient stability, and all four architectures' bounded checks

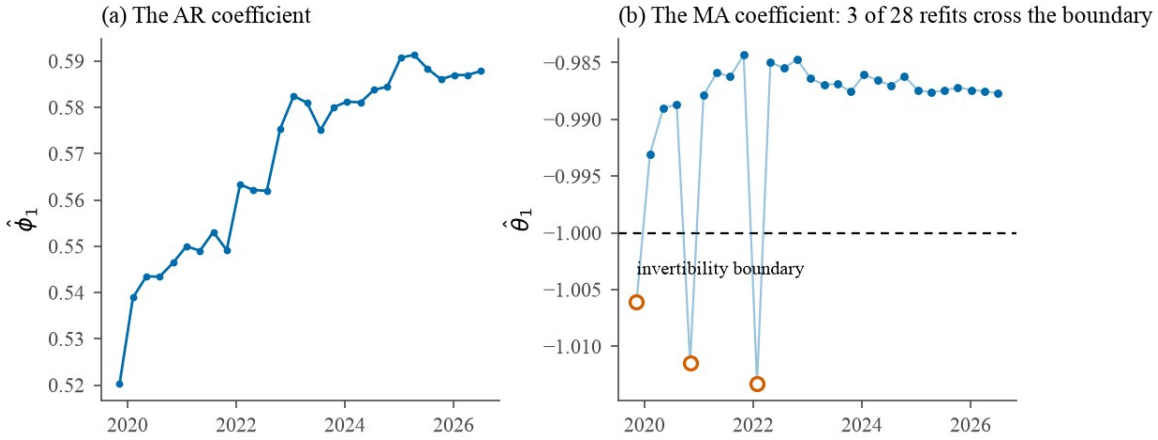


Figure 3.6. Both coefficients across all 28 refits; the three open points cross the invertibility boundary.

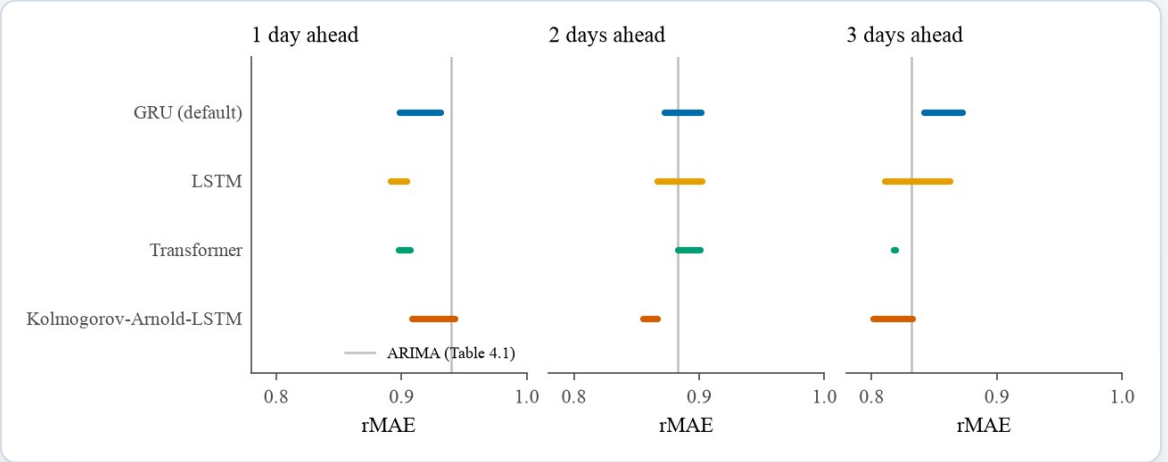


Figure 5.2. Bounded checks, ranges over three seeds, against ARIMA's rolling-origin rMAE.

BACKUP

B5. RQ2 at all three horizons: the hit-rate advantage of forecast-then-threshold

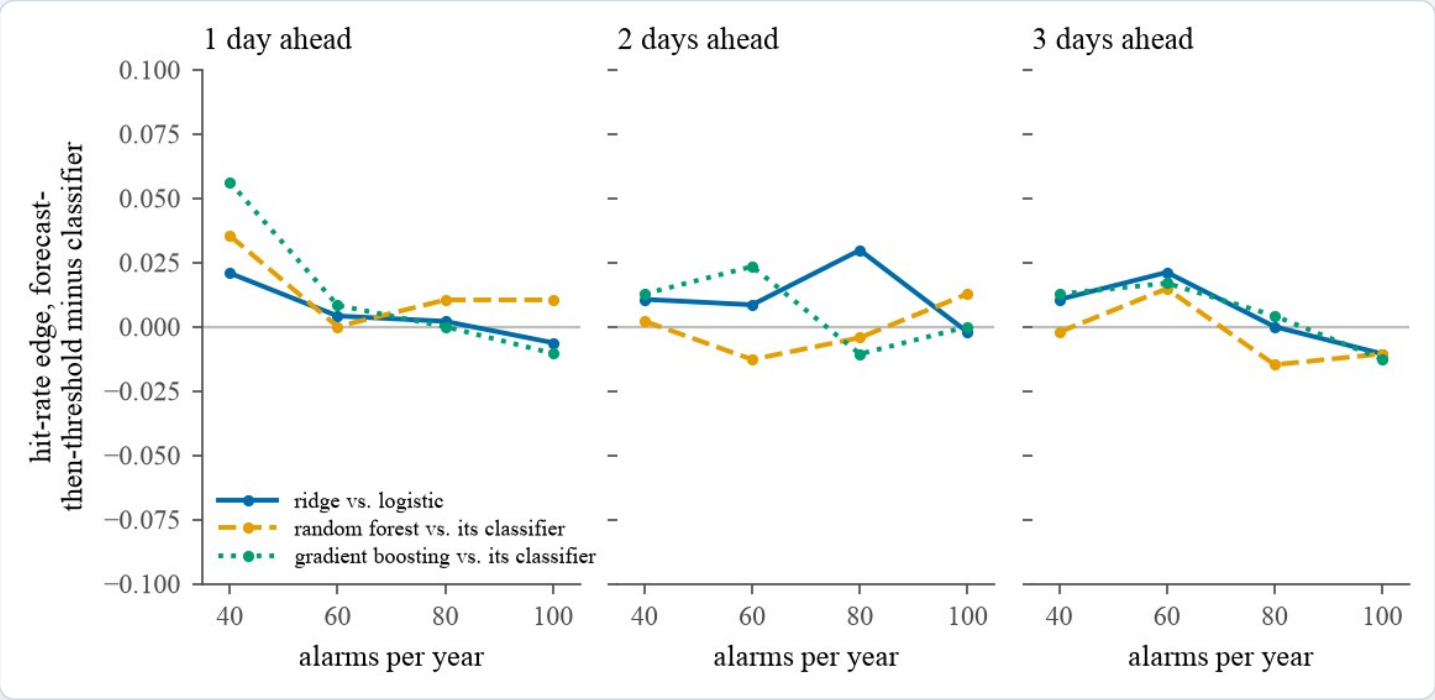


Figure 4.4. Above the zero line forecast-then-threshold wins; below it the classifier does. One line per matched pair.

Forecast-then-threshold wins 8 of 12 matched comparisons at one day ahead, 7 of 12 at two days and 6 of 12 at three: an edge that narrows rather than changing hands.